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Quiz 3: BigQuery

100% (100%)

Done Apr 21, 11:59 PM '27

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100% correct, 0% wrong • You achieved 100% • To pass you need at least 80%. We keep your latest score.

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1. In BigQuery's serverless architecture, query processing capacity is tightly coupled with the amount of data you have stored. To improve query performance, you must first increase your storage capacity.

False

True

Correct

That's correct! BigQuery's architecture fundamentally separates compute from storage. This is a core concept. Query processing is handled by the Distributed Engine, which abstracts and isolates compute resources independently and independently of the storage layer. You do not need to provision or scale storage to get more query power. This separation allows BigQuery to handle petabyte-scale queries without requiring users to manage underlying infrastructure.

2. A development team is testing the operational data for a mobile application in a BigQuery instance to ensure the latency needs and write for the app. A business analyst needs to run an ad hoc SQL query on the live data to validate a recent trend. Which is the most efficient and recommended way to achieve this without impacting the mobile application's performance?

Export the data from BigQuery to a CSV file in Cloud Storage, and then load that CSV file into BigQuery.

Load the dataset from an external cloud storage to the production BigQuery instance and have them use the C3I command-line tool to refresh the data.

Create an external table in BigQuery that points to the BigQuery instance, and have the analyst run their query against that external table using a "Table Store" profile.

Implement a workflow pipeline to read all the data from BigQuery and write it into a new, native BigQuery table for the analyst's query.

Correct

That's right! This is the exact use case for the BigQuery and BigQuery integration. Creating an external table allows BigQuery to query the data stored in the BigQuery instance using the C3I. Using the C3I tool is a critical part that ensures performance isolation. It runs the analyst query using dedicated, on-demand compute resources, which guarantees that the heavy analytical workload will not interfere with the low-latency traffic from the production mobile applications.

3. A data analyst complains that their queries on a large, multi-year historical sales database are slow and costly. The queries often filter by the transaction date and group by product category. Which of the following BigQuery features could be used to improve query performance and reduce costs for this specific workload? (Select all that apply.)

Using the BigQuery Storage Write API

Clustering the table by product category

Correct

That's correct! After partitioning by date, clustering by product, categories would physically co-locate data for the same category within each partition. This improves the performance of queries that filter or group by category. As BigQuery can more efficiently read the relevant blocks of data.

Creating a materialized view for common aggregations

Correct

That's right! For running common aggregations (grouping by category), a materialized view could pre-compute and store those results. Queries that match the materialized view's logic would read from the much smaller, pre-aggregated table instead of the raw data, saving them significant time and expense.

Partitioning the table by transaction date

Correct

That's correct! Partitioning the table by date (for example, daily or monthly) is the most effective optimization here. When queries filter by the transaction date, BigQuery can prune partitions, meaning it only scans the data in the relevant date partitions, essentially reducing the amount of data processed, which lowers cost and improves speed.

4. A social media company uses a single, massive BigQuery table to store all user events (likes, posts, comments), which partitioned by day. The Data Science team needs to run complex machine learning model, concurrent dashboard queries that typically look at the last 7 days of data. The analytics dashboards are running for the whole time for the BigQuery team. What is the best way to ensure the Product Analytics team's dashboards remain fast and responsive without impacting the Data Science team's work?

Switch BigQuery from the on-demand pricing model to a flat rate model and create separate memberships for the Data Science and Product Analytics teams.

Ask the Data Science team to only run their heavy queries at night when the Product Analytics team is offline.

Use a BigQuery BI Engine reservation to accelerate the dashboards.

Split the single table into two separate tables: one for historical data and one for the last 7 days.

Correct

That's right! Flat rate reservations are designed specifically for this scenario. By purchasing a reserved amount of data capacity for BigQuery, you can offload and storing large-scale computations, you can offload one part of data to the Data Science team (for their heavy, long-running jobs) and another part to the Analytics team for their short, high-concurrency dashboard queries. This guarantees that a surge in demand from one team does not consume the compute resources needed by the other, ensuring consistent performance for both.